

Contestant Number: _____

Time: _____

Rank: _____



VISUAL BASIC C# PROGRAMMING (330)

STATE – 2020

Production Portion:

Program 1: PDF Manager Application _____ (450 points)

TOTAL POINTS _____ ***(450 points)***

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than ten (10) minutes orientation

No more than 90 minutes testing time

No more than ten (10) minutes wrap-up

Property of Business Professionals of America.
May be reproduced only for use in the Business Professionals of America
Workplace Skills Assessment Program competition.

PDF Manager Application

Textbooks are becoming less and less common in our education system, with more teachers and students preferring the use of e-books (PDFs for simplicity). For this exercise, you will create a Visual Basic or C# Windows Form Application that is suitable for managing user login, the PDFs a user owns, viewing those PDFs, and purchasing additional PDFs. XML Serialization will be used for a database to store both PDF and user information.

Requirements:

1. You must create a C#/VB Windows Form Application named CS_330_ContestantNumber, where ContestantNumber is your BPA assigned contestant number (including dashes). For example, CS_330_01_2345_6789. If you are using VB, use VB instead of CS in the application name.
2. Your contestant number must appear as a comment at the top of the main source code file.
3. On application start, a login form shall appear.
 - a. This must be done using a different form than the main application.
 - b. When started, both the user and pdf databases shall be loaded from files “pdfs.xml” and “users.xml”. The databases can be represented as lists of User and Pdf objects.
 - c. If a proper entry is made (defined as username and password matching an existing user’s information), the Login form shall disappear and the main application shall launch.
 - d. The login form shall close after the main application closes.
 - e. The login form, with proper entry, shall pass both the user and pdf databases as well as the integer index the logged in user can be found at.
4. The login form must not have minimize or maximize buttons in the Form Handle.
5. The login form must have text “Login” for its title.
6. The login form must have textboxes for Username and Password entry, named “userBox” and “passBox” respectively and with suitable labels named “userLbl” and “passLbl”.
 - a. “userBox” shall have focus immediately upon form launch and pressing tab shall shift focus to “passBox”.
 - b. Pressing enter shall invoke a button “loginBtn” with text “Login” to check the user’s entry.
7. If invalid user information is entered, display a warning with message “Invalid login.”
8. Two classes must be created – Pdf and User – to store the information from the two XML databases.
 - a. The Pdf class shall have integer Id, string File, and double Cost public properties
 - b. The User class shall have string Name, string Password, integer list Pdfs, and double Money public properties
 - i. The Pdfs integer list shall be used to relate to a Pdf objects Id field and shall be used to mark which Pdfs a user has access to.

- c. The User class shall have static functions for saving and loading, while the Pdf class shall only have a static load function, named “SaveUsers”, “LoadUsers”, and “LoadPdfs” respectively.
 - i. The Save functions shall take in the current database (list of objects)
 - ii. The Load functions shall attempt to load in the database file, but if not present shall create a new database file with sample data. The loaded database shall then be returned. The specifications for sample data are provided in the included pdfs.xml and users.xml.
 - iii. Both sets of functions shall have suitable error messages when either fails.
 - d. Suitable constructors shall be created for both classes.
 - i. An empty constructor.
 - ii. A constructor that accepts values for each public property.
 - e. The User class shall have a “PurchasePdf” public function with a Pdf object as a parameter.
 - i. The function shall check if the user has enough money for the selected pdf, and if so, deduct the cost while adding the pdf’s Id field to the Pdfs integer list of the user.
 - ii. Shall return a bool value indicating if purchase was successful.
9. Loading and saving of User and Pdf databases shall be done using XML serialization. The code for performing generic XML serialization is included in the provided Serialization class and shall be used by the “LoadUsers” and “LoadPdfs” functions.
10. Main Form on close must ask for confirmation to close, with the Title of the confirmation box being “Confirm” and the text being “Are you sure you wish to exit the PDF Manager?”
- a. Login form shall be allowed to close without asking confirmation.
 - b. When the main form successfully closes, the SaveUsers function should be called to update the database file.
11. The main form must have text “PDF Manager” for its title.
12. The main form must have a tab control containing three tabs.
- a. The tab control must be named “pdfManagerTabs”, with Dock set to Fill.
 - b. The first tab must be named “pdfSelectPage” with text “My PDFs”. It must contain a button with text “Select PDF” and named “selectPdfBtn” with dock set to Bottom as well as a ListBox named “ownedPdfList” with dock set to Fill.
 - c. The second tab must be named “currentPdfPage” with text “Current PDF”. It must contain a WebBrowser named “pdfViewer” with dock set to Fill.
 - d. The third tab must be named “newPdfPage” with text “Get New PDFs”. It must contain a button with text “Purchase PDF” and named “buyPdfBtn” with dock set to Bottom as well as a ListBox named “buyablePdfList” with dock set to Fill.
13. When the main form is constructed, the current user index shall be used in conjunction with the two passed databases to populate the buyablePdfList and ownedPdfList controls.
- a. Any pdfs with Ids matching what is in the current user’s Pdfs list shall be added to the ownedPdfList control, otherwise it shall be added to the buyablePdfList control.

- b. This should be done using a suitable function that also clears these two lists prior to population.
 - c. When a pdf is purchased by the current user, this action should be performed again to refresh the list.
- 14. The sample PDF files shall be stored in the same directory as the application. **The contestant will need to move them to this directory after the application is built the first time.**
- 15. When the selectPdfBtn is pressed, the following occurs:
 - a. It is verified a single item is selected in the ownedPdfList.
 - b. It is verified the selected file exists.
 - c. If both conditions are present, the pdfViewer WebBrowser control navigates to that file to load the file, and the TabControl switches the selected tab to the viewing page.
 - d. A failure at any of these steps shall result in an error message.
- 16. When the buyPdfBtn is pressed, the following occurs:
 - a. It is verified a single item is selected in the buyablePdfList.
 - b. It is verified the selected file exists.
 - c. The Pdf object is retrieved from the database using the string stored by the buyablePdfList selected item.
 - d. The current user attempts to purchase the item using the PurchasePdf function.
 - e. A failure at any of these steps shall result in an error message.

Sample Pictures:

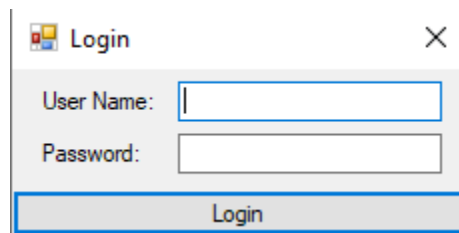


Figure 1: The login form

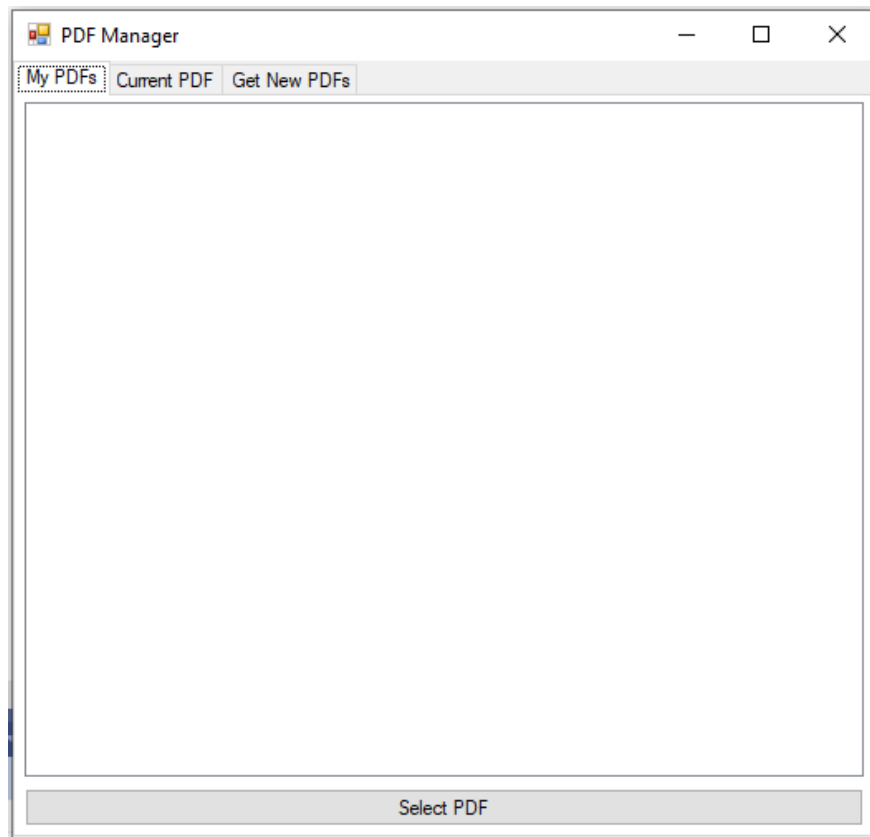


Figure 2: The PDF Manager form with no purchased PDFs.

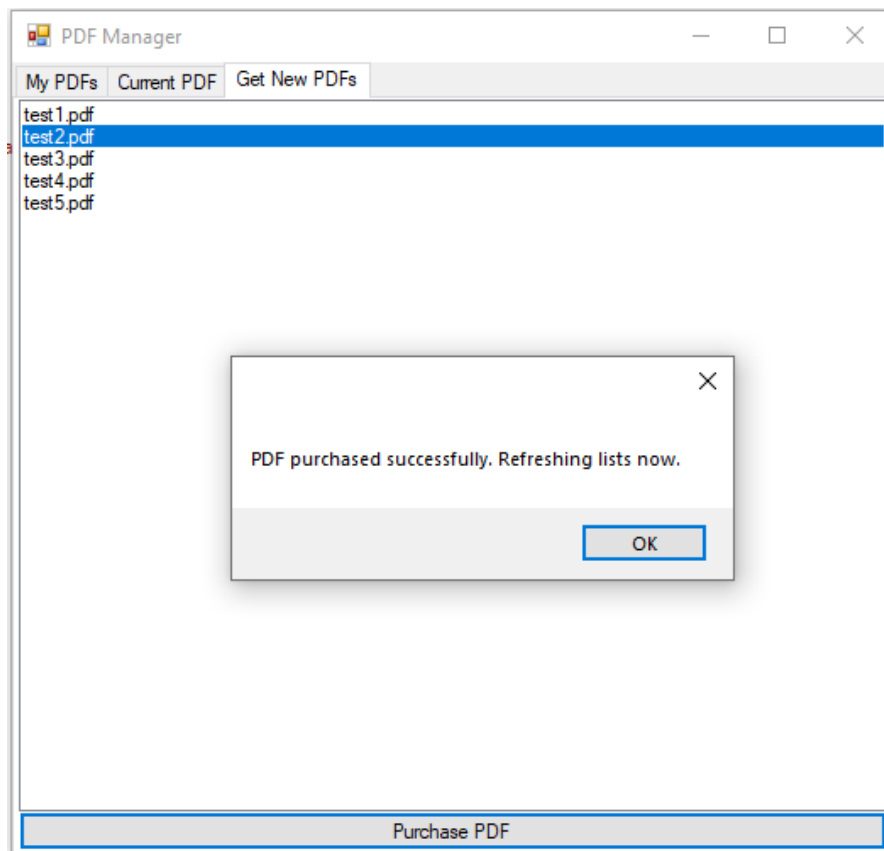


Figure 3: Successful purchasing of test2.pdf. After hitting OK, test2.pdf is removed from this list.

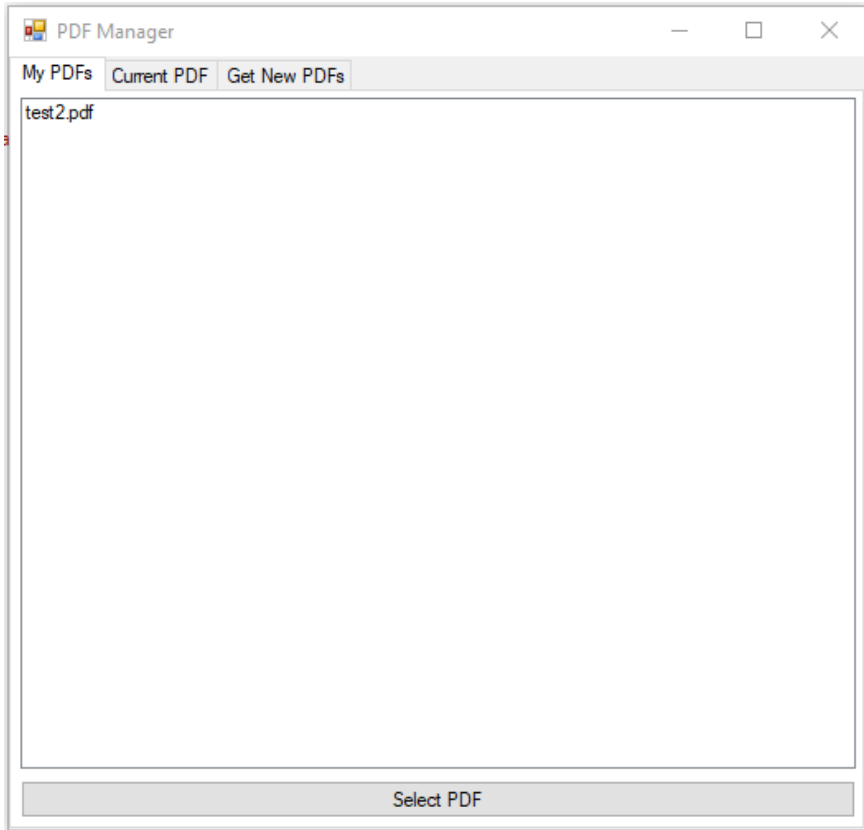


Figure 4: Successful purchasing of test2.pdf, it is shown in My PDFs.

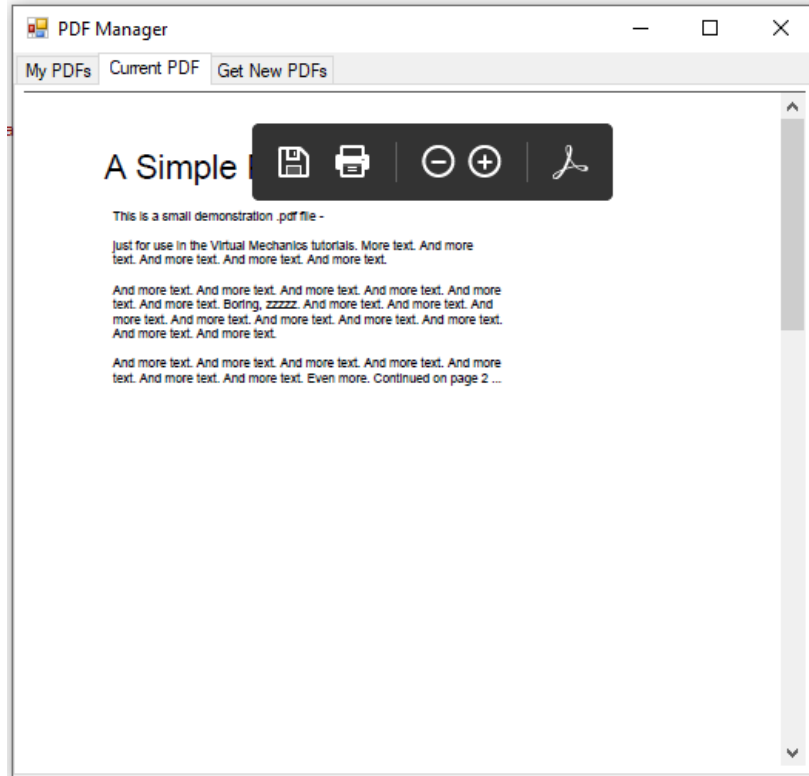


Figure 5: After pressing Select PDF, the PDF is loaded using the pdfViewer WebBrowser and the active tab page is switched to Current PDF.

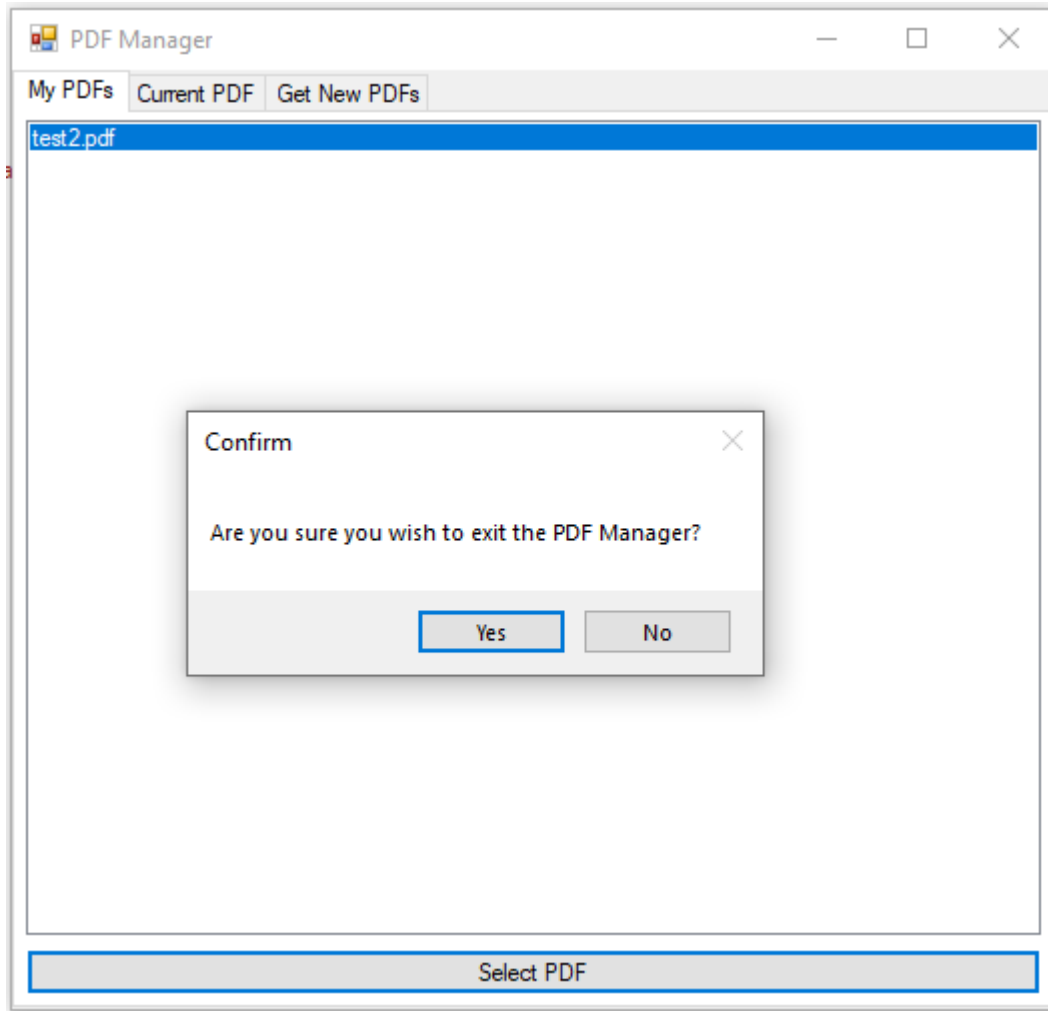


Figure 6: The confirmation window. This should be opened when Exit is pressed. If “Yes” is clicked, the application should close, otherwise return to the main form.

You will have 90 minutes to complete your work.

Your name and/or school name should NOT appear on any work you submit for grading.

Submit a copy your entire solution/project to the flash drive provided. You must submit your entire solution/project so that the graders may open your project to review the source code and/or build and execute your solution/project. **Submissions that do not contain source code will not be graded.**

Development Standards

- Standard name prefixes must be utilized for variables.
- All subroutines, functions, and methods must be documented with comments explaining the purpose of the method, the input parameters (if any), and the output (if any).

Your application will be graded on the following criteria:

Solution and Project

The project is present on the flash drive _____ 10 pts
The project is named according to the naming conventions _____ 10 pts

Program Execution

Code copied to USB drive and program runs from USB _____ 20 pts

If the program does not execute, then the remaining items in this section receive a score of zero.

The main form asks for confirmation on close, login does not _____ 15 pts
Database state is saved on closing main form _____ 20 pts
Database state is loaded on opening login form from pdfs.xml and users.xml _____ 20 pts
Textboxes are present to enter user information in login form _____ 10 pts
Login successfully pulls correct user information from database _____ 25 pts
Main form is initialized with current user and database after login _____ 25 pts
Login form properly disappears after successful login _____ 10 pts
Main form has tab control with suitable pages _____ 20 pts
Each page has specified controls with similar layout _____ 20 pts
On main form loading and purchase of Pdf, PDF lists are refreshed _____ 25 pts
My PDFs page is refreshed with owned PDFs _____ 10 pts
Get New PDFs page is refreshed only with not owned PDFs _____ 10 pts
Selecting in My PDFs results in transition to Current PDF and loading the file _____ 30 pts
PDF purchasing if user has enough money and adds PDF to owned list _____ 20 pts
Error messages are displayed according to specifications _____ 20 pts

Source Code Review

Code is commented at the top, for each function, and as needed _____ 10 pts
Code uses reasonable and consistent variable naming conventions _____ 10 pts
Application makes use of two separate forms _____ 20 pts
Provided Serialization class is used for XML Serialization _____ 20 pts
User class is created and has specified variables and functions _____ 10 pts
Pdf class is created and has specified variables and functions _____ 10 pts
Suitable constructors are present for both User and Pdf classes _____ 10 pts
Controls are named as described in requirements _____ 20 pts
WebBrowser control is used to load current PDF _____ 20 pts

Total Points: 450 pts